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Thermodynamic modeling of the gadolinium–germanium system

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Abstract

A thermodynamic modeling of the Gd–Ge system was carried out and a consistent set of thermodynamic parameters were obtained. The liquid is modeled as substitutional solution, the α-Gd (bcc-A2), β-Gd (hcp-A3) and Ge (diamond-A4) are considered to be pure stoichiometric phases, and all intermediate compounds are modeled as stoichiometric phases. The currently accepted thermodynamic description of pure gadolinium [1] was presently amended in this work.
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1. Introduction

Magnetic refrigeration, which works by utilizing the heat release or absorption caused by the magnetic entropy change (ΔS_M) due to a magnetic field change (ΔH) [2], is becoming a promising technology to replace the conventional gas-compression expansion technique. Giant ΔS_M has recently been observed in the series of $Gd_5(Ge_{1-x}Si_x)_4$ ($0.43 \le x \le 0.52$) intermetallic compounds [3,4], which is of high interest nowadays because of its potential application in the magnetocaloric material. Accurate knowledge of the Gd–Ge–Si ternary system, as well as its sub-systems becomes necessary. The present work is to provide a self-consistent thermodynamic description of the sub-binary Gd–Ge system.

2. Evaluation of experimental data

2.1. Phase equilibria and crystallographic data

The crystallographic data and phase equilibria data of the Gd–Ge system available in the literature up to 1986 had been critically reviewed by Gokhale and Abbaschian [5]. The only contribution to phase equilibria in the Gd–Ge system owes to Eremenko et al. [6], who thoroughly investigated this system using metallography, differential thermal analysis (DTA) and X-ray diffraction (XRD), and established a Gd–Ge phase diagram in the entire composition range: the liquidus was reliably determined over the entire composition range; the mutual terminal solid solubilities of the pure components were not reported, but are probably very small; the intermediate compounds, Gd_5Ge_3, Gd_5Ge_4, GdGe, α-Gd_2Ge_3, β-Gd_2Ge_3, γ-Gd_2Ge_3, Gd_3Ge_5 and $GdGe_{2.57}$, along with their melting temperatures and characteristics were ascertained.

The intermediate compound Gd_5Ge_3 (space group: $P6_3/mcm$; prototype: Mn_5Si_3), richest in Gd, had been observed in a large temperature range from 773 to 1573 K by several groups of investigators [6–14]. It melts peritectically at 2063 K and a eutectic equilibrium ($L \leftrightarrow$ α-Gd + Gd_5Ge_3) associated with it occurs at 1386 K and 11 at.% Ge [6]. The Gd_5Ge_4 phase (space group: *Pnma*; prototype: Sm_5Ge_4), has been determined by Smith et al. [15,16] using single crystal method and was confirmed by later investigators [6,17]. It forms peritectically at 1963 K [6]. The GdGe phase (space group: *Cmcm*; prototype: CrB) has been determined by a lot of investigators [6,9,14,18–22], and it forms peritectically at 1843 K [6]. The $GdGe_{2.57}$ phase (space group: $C222_1$) was determined by Eremenko et al. [6] using XRD and DTA. It forms peritectically at 1163 K and decomposes eutectoidally at 1123 K, and a eutectic equilibrium ($L \leftrightarrow GdGe_{2.57}$ + Ge) associated with it occurs at 1133 K and 85 at.% Ge [6]. All these compounds were estimated to be stoichiometric since no homogeneity range was observed for them [5,6].

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In the small composition range from 59 to 62.5 at.% Ge, two compounds with nominal formula of Gd_2Ge_3 and Gd_3Ge_5 had been established by Eremenko et al. [6]. They experimentally detected a AlB_2-type compound (space group: *P*6/*mmm*) at 60 at.% Ge and 973 K. In order to interpret their DTA observations, Eremenko et al. [6] proposed three allotropic forms for Gd_2Ge_3: the high-temperature γ-Gd_2Ge_3 forms peritectically at 1688 K; γ-Gd_2Ge_3 transforms to β-Gd_2Ge_3 at temperatures from 1483 to 1343 K decreasing with the Ge content increasing from 59 to 61 at.%; β-Gd_2Ge_3 transforms to the low-temperature α-Gd_2Ge_3 at 1106 K; the two high-temperature structures were not determined. The DTA observations at 1483 and 1343 K [6] are considered to be reliable, but need to be reinterpreted, since the allotropic transition γ-Gd_2Ge_3 ↔ β-Gd_2Ge_3 lacks experimental evidence and such a transition is difficult to be modeled. In the present work, the three forms of Gd_2Ge_3 were assumed to be stoichiometric, α-Gd_2Ge_3 and β-Gd_2Ge_3 at 59 at.% Ge and γ-Gd_2Ge_3 at 60 at.% Ge, and therefore two invariant equilibria occur, γ-Gd_2Ge_3 + GdGe ↔ β-Gd_2G_3 at 1483 K and γ-Gd_2Ge_3 ↔ β-Gd_2Ge_3 + L at 1343 K. Gd_3Ge_5 has a $ThSi_2$-type structure (space group: $I4_1/amd$) and forms peritectoidally at 1081 K [6]. Gokhale and Abbaschian [5] proposed an allotropic transformation (α-Gd_3Ge_5 ↔ β-Gd_3Ge_5 below 600 °C) by reinterpreting the previous experimental findings of the α-$GdSi_2$-type "$GdGe_{2-n}$" phase from Gladyshevskii and Burnashkova [23], and estimated a homogeneity range from 61.64 to 62.55 at.% Ge for Gd_3Ge_5.

Following the previous assessment [5], the melting/allotropic transformation temperatures of the pure Gd were changed to conform to the experimental values from Gschneidner and Calderwood [24].

2.2. Thermodynamic data

By means of emf, Batalin et al. [25] determined the Gibbs energy of formation of $GdGe_{1.63}$ (Gd_3Ge_5) as a function dependent of temperature: $\Delta G_f = -210965 + 13.71T$ J/mol-formula (T in K). From it, the enthalpy of formation of Gd_3Ge_5 can be derived to be about −80.21 kJ/mol-atoms. The same group afterwards determined the Gibbs free energy, enthalpy and entropy of formation of the stable compounds also by the emf method [26]. The enthalpies of formation (kJ/mol-atoms) were reported to be −82.9 ± 3.4 for Gd_3Ge_5, −87.7 ± 3.6 for Gd_2Ge_3, −105.4 ± 5.0 for GdGe, −115.3 ± 5.5 for Gd_5Ge_4 and −112.8 ± 7.0 for Gd_5Ge_3. Using direct-synthesis calorimetry, Meschel and Kleppa [27] determined the standard enthalpy of formation of Gd_5Ge_3 to be −82.0 ± 2.6 kJ/mol-atoms at 1473 ± 2 K. Usenko et al. [28] recently measured the standard molar enthalpies of formation (kJ/mol-atoms) using solution drop calorimetry: −65.0 ± 4.1 for Gd_3Ge_5, −70.7 ± 4.1 for Gd_2Ge_3, −79.6 ± 3.5 for GdGe, −79.5 ± 3.1 for Gd_5Ge_4 and −83.1 ± 3.4 for Gd_5Ge_3. The results somewhat differ from the data obtained by the emf method in Refs. [25,26], but agree well with the direct calorimetric measurement of Meschel and Kleppa [27]. The dissolution experiments had been repeated using different solvents (Al, Cu and Sn) and the reliability of the obtained results was confirmed [28]. Therefore, the data of Usenko et al. [28], as well as that of Meschel and Kleppa [27], were utilized in this modeling.

3. Thermodynamic modeling

3.1. Unary phases

The Gibbs energy function, $G_i^{0,\phi}(T) = G_i^{\phi}(T) - H_i^{SER}$, for the element i (i = Gd, Ge) in Φ phase (Φ = β-Gd (bcc-A2), α-Gd (hcp-A3), Ge (diamond) and liquid) is given by

$$G_i^{0,\phi}(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

H_i^{SER} is the molar enthalpy of the so-called "standard element reference" (SER), *i.e.* the enthalpies of the pure elements in their defined reference phase at 298.15 K and 1 bar, T is the absolute temperature. The β-Gd, α-Gd and Ge are considered to be pure stoichiometric phases since the mutual terminal solid solubilities of the pure components can be negligible.

The thermodynamic description for pure Ge is taken from the SGTE compilation by Dinsdale [29], while that for Gd from the same compilation was not adopted because it does not conform to the currently accepted melting/allotropic transformation temperatures of Gd [24]. Shen et al. [1] recently assessed the Gibbs energies of bcc-A2, hcp-A3 and liquid of Gd, based on the analysis and evaluation of the thermodynamic information including heat capacity, enthalpy, relative magnetic parameter, solid transition temperature, melting point, solid transition enthalpy and melting enthalpy. For hcp-A3 Gd, the Gibbs energy had been described by splitting it into a nonmagnetic contribution (${}^0G^{nmg}$) and a magnetic one (${}^{\Delta}G^{mg}$). The nonmagnetic contribution is described by Eq. (1), while the magnetic contribution by the Hillert–Jarl–Inden model [30]. The thermodynamic description of Gd from Shen et al. [1] was thus utilized in this work but with a slight modification, which would be discussed in the following.

Appendix A presents the thermodynamic description of Gd from Shen et al. [1], which is dependent of temperature (T) and expressed as different polynomials in different temperature ranges. There are several critical temperatures, at which the functions are discontinuous resulting from the description although the Gibbs energies are continuous in nature. One of these temperatures corresponds to the allotropic transition between α-Gd and β-Gd occurs at 1508.15 K [24]. Unfortunately, it was found that the α-Gd ↔ β-Gd transition cannot be computed using this description.

In order to reveal the reason for this problem, the Gibbs energy of β-Gd relative to the reference state (α-Gd) was calculated, which was denoted as Gmr(bcc) in the following discussion. It is worth noting that there is a magnetic contribution to the Gibbs energy of α-Gd, although it is slight. Fig. 1a presents the calculated curve in the small temperature range around 1508.15 K and it shows that there exists a discontinuity at 1508.15 K. An auxiliary horizontal line was drawn at Gmr(bcc) = 0. It was illustrated that the curve of Gmr(bcc) cannot yield a value of zero, which implies that the Gibbs energy

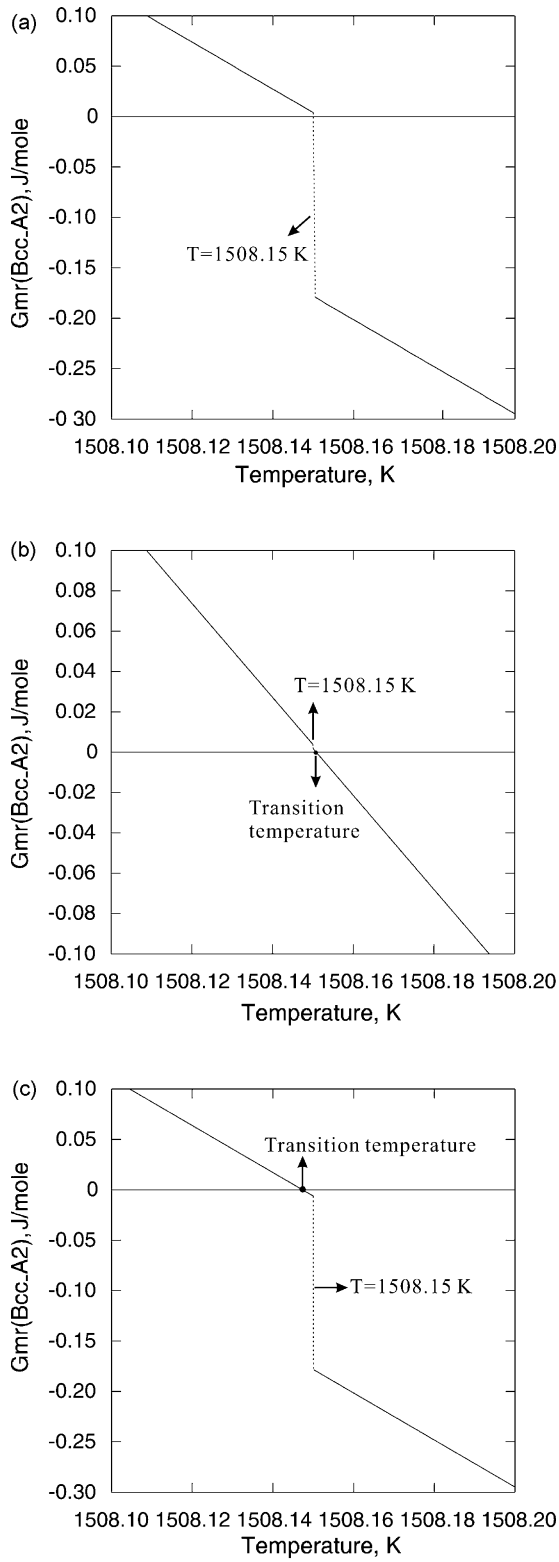

Fig. 1. The Gibbs energy curve of β-Gd (bcc) relative to α-Gd (hcp) around 1508.15 K, according to (a) the original thermodynamic description of Gd from Shen et al. [1], (b) modification A and (c) modification B.

of α-Gd cannot be equal to that of β-Gd. That is the reason why the α-Gd ↔ β-Gd transition cannot be computed.

The solution for this problem is to make Gmr(bcc) zero at a temperature very close to 1508.15 K. It can be achieved by adjusting the Gibbs energy functions of bcc or hcp very slightly. Although more specific solutions are possible, two alternative modifications are provided in this work, as given in Appendix A. Modification A yields a transition temperature slightly higher than 1508.15 K, as illustrated in Fig. 1(b). Modification B yields a transition temperature slightly lower than 1508.15 K, as illustrated in Fig. 1(c). The former is preferable and finally adopted in the present modeling since it provides a relatively more slippery Gibbs energy function for β-Gd.

Resulting from this modification, the transition α-Gd ↔ β-Gd can be computed. It is worth noting that the transition temperature does not equal to 1508.15 K any more, however, the deviation is negligible. The modified description of β-Gd is proposed to be used for the thermodynamic modeling of the systems containing Gd.

3.2. Liquid

The liquid phase is treated as a substitutional solution and its Gibbs energy is expressed as

$$G_{\mathrm{m}}^{\mathrm{liquid}} = x_{\mathrm{Gd}}{}^{0}G_{\mathrm{Gd}}^{\mathrm{liquid}} + x_{\mathrm{Ge}}{}^{0}G_{\mathrm{Ge}}^{\mathrm{liquid}} + RT(x_{\mathrm{Gd}}\ \ln\ x_{\mathrm{Gd}} + x_{\mathrm{Ge}}\ \ln\ x_{\mathrm{Ge}}) + {}^{\mathrm{ex}}G_{\mathrm{m}}^{\mathrm{liquid}} \quad (2)$$

The excess Gibbs energy, ${}^{\mathrm{ex}}G_{\mathrm{m}}^{\mathrm{liquid}}$, is described by the Redlich–Kister polynomial [31]:

$${}^{\mathrm{ex}}G_{\mathrm{m}}^{\mathrm{liquid}} = x_{\mathrm{Gd}}x_{\mathrm{Ge}}\sum_{i=0}^{n}{}^{i}L^{\mathrm{liquid}}(x_{\mathrm{Gd}} - x_{\mathrm{Ge}})^{i} \quad (3)$$

The interaction parameters between components Gd and Ge, ${}^{i}L^{\mathrm{liquid}}$, is taken as a linear function of temperature:

$${}^{i}L^{\mathrm{liquid}} = a_i + b_i T \quad (4)$$

3.3. Intermediate phases

All intermediate phases Gd_5G_3, Gd_5Ge_4, GdGe, α-Gd_2Ge_3, β-Gd_2Ge_3, γ-Gd_2Ge_3, α-Gd_3Ge_5, β-Gd_3Ge_5 and $GdGe_{2.57}$ are treated as stoichiometric phases. The Gibbs energy of each phase per mol-atoms is described by the following formula:

$$G^{\mathrm{Gd}_x\mathrm{Ge}_y}(T) = \frac{x}{x+y}{}^{0}G_{\mathrm{Gd}}^{\mathrm{hcp\text{-}A3}}(T) + \frac{y}{x+y}{}^{0}G_{\mathrm{Ge}}^{\mathrm{diamond\text{-}A4}}(T) + A + BT \quad (5)$$

where A and B are parameters to be optimized. For intermediate phases, starting values for coefficients A can be estimated according to the corresponding experimental standard enthalpies of formation [28]. The coefficients were then evaluated based on the starting values by taking into account both phase equilibria and thermodynamic data. Coefficients B were optional and could be utilized in the optimization if necessary.

4. Result and discussion

Evaluation of the model parameters is attained by recurrent run of the PARROT program, which uses a least-square procedure [32]. In the procedure of optimization, each piece of experimental data is given a certain weight. The weights were changed systematically during the assessment until all utilized experimental data were accounted for within the experimental uncertainties claimed in the literature or evaluated in this work.

4.1. Phase diagram

Table 1 presents the finally evaluated thermodynamic parameters of each phase in the Gd–Ge system. The calculated Gd–Ge phase diagram are presented in Fig. 2a and compared with experimental data of Eremenko et al. [6] in Fig. 2b, along with the Ge-rich portion shown on an expended scale in Fig. 2c. All phase diagram data are well reproduced in the present calculation. Table 2 compares all calculated and measured invariant equilibria. The differences between the measured and calculated invariant transition temperatures are less than 1 K.

4.2. On the allotropic transformation α-Gd_3Ge_5 ↔ β-Gd_3Ge_5

The allotropic transformation α-Gd_3Ge_5 ↔ β-Gd_3Ge_5 has been estimated to be below 873 K by Gokhale and Abbaschian [5]. Its existence needs further experimental evidences. The preliminary thermodynamic optimization had been carried out without taking into account the allotropic transformation, i.e., by excluding α-Gd_3Ge_5. The coefficient A for β-Gd_3Ge_5 was evaluated to be −65,305 whereas the coefficient B was not used. However, the model and parameter for α-Gd_3Ge_5 could be easily

Table 1
Evaluated thermodynamic parameters of each phase in the Gd–Ge system

Liquid: $(Gd,Ge)_1$

$^0L^{liq}_{Gd,Ge} = -239,589 + 8.37614T$

$^1L^{liq}_{Gd,Ge} = -8598$

$^2L^{liq}_{Gd,Ge} = +60,318$

α-Gd: Gd (hcp-A3)

β-Gd: Gd (bcc-A2)

Gd_5Ge_3: $Gd_{0.625}Ge_{0.375}$

$^0G^{Gd_5Ge_3}_{Gd:Ge} - 0.625\,^0G^{hcp}_{Gd} - 0.375\,^0G^{dia}_{Ge} = -81,104 + 2.82415T$

Gd_5Ge_4: $Gd_{0.55555}Ge_{0.44445}$

$^0G^{Gd_5Ge4}_{Gd:Ge} - 0.55555\,^0G^{hcp}_{Gd} - 0.44444\,^0G^{dia}_{Ge} = -81,512 + 1.59572T$

GdGe: $Gd_{0.5}Ge_{0.5}$

$^0G^{GdGe}_{Gd:Ge} - 0.5\,^0G^{hcp}_{Gd} - 0.5\,^0G^{dia}_{Ge} = -82,179 + 2.31689T$

α-Gd_2Ge_3: $Gd_{0.41}Ge_{0.59}$

$^0G^{Gd_2Ge_3}_{Gd:Ge} - 0.41\,^0G^{hcp}_{Gd} - 0.59\,^0G^{dia}_{Ge} = -71,076 - 0.30000T$

β-Gd_2Ge_3: $Gd_{0.41}Ge_{0.59}$

$^0G^{Gd_5Ge_3}_{Gd:Ge} - 0.41\,^0G^{hcp}_{Gd} - 0.59\,^0G^{dia}_{Ge} = -71,001 - 0.36781T$

γ-Gd_2Ge_3: $Gd_{0.40}Ge_{0.60}$

$^0G^{Gd_2Ge_3}_{Gd:Ge} - 0.4\,^0G^{hcp}_{Gd} - 0.6\,^0G^{dia}_{Ge} = -63,198 - 5.08952T$

α-Gd_3Ge_5: $Gd_{0.375}Ge_{0.625}$

$^0G^{Gd_3Ge_5}_{Gd:Ge} - 0.375\,^0G^{hcp}_{Gd} - 0.625\,^0G^{dia}_{Ge} = -65,741 + 0.50000T$

β-Gd_3Ge_5: $Gd_{0.375}Ge_{0.625}$

$^0G^{Gd_3Ge_5}_{Gd:Ge} - 0.375\,^0G^{hcp}_{Gd} - 0.625\,^0G^{dia}_{Ge} = -65,305$

$GdGe_{2.57}$: $Gd_{0.28011}Ge_{0.71989}$

$^0G^{GdGe_{2.57}}_{Gd:Ge} - 0.28011\,^0G^{hcp}_{Gd} - 0.71989G^{dia}_{Ge} = -34,599 - 12.63803T$

Ge: Ge (diamond-A4)

Note: Gibbs energy in J/mol-atoms and T in Kelvin.

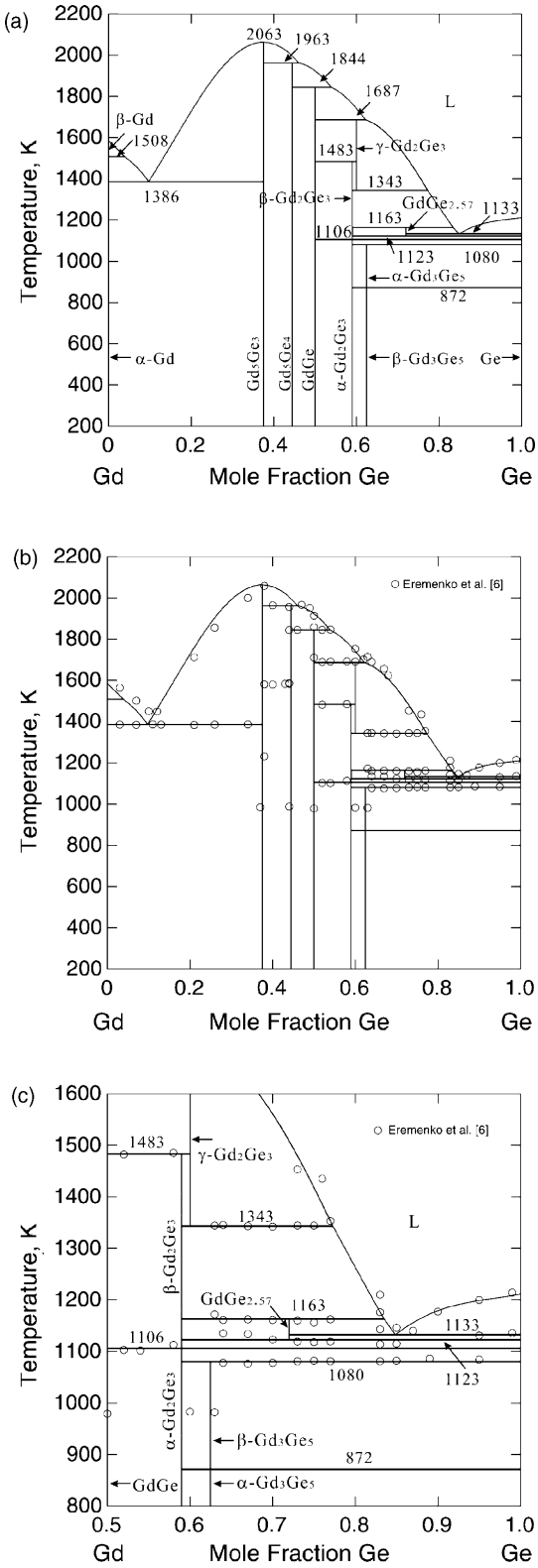

Fig. 2. Calculated Gd–Ge phase diagram in this work (a) labeled with phases and the transition temperatures, (b) compared with experimental data of Eremenko et al. [6] and (c) in the range from 50 to 100 at.% Ge on an expanded scale.

Table 2
Calculated and measured invariant equilibria in the Gd–Ge system

Reaction	T (K)	Comment	Reference	Calculated
$L \leftrightarrow \alpha\text{-}Gd + Gd_5Ge_3$	1386	Measured	[6]	1386
$L \leftrightarrow Gd_5Ge_3$	2063	Measured	[6]	2063
$L + Gd_5Ge_3 \leftrightarrow Gd_5Ge_4$	1963	Measured	[6]	1963
$L + Gd_5Ge_4 \leftrightarrow GdGe$	1843	Measured	[6]	1844
$L + GdGe \leftrightarrow \gamma\text{-}Gd_2Ge_3$	1688	Measured	[6]	1687
$\gamma\text{-}Gd_2Ge_3 + GdGe \leftrightarrow \beta\text{-}Gd_2Ge_3$	1483	Measured	[6]	1483
$\gamma\text{-}Gd_2Ge_3 \leftrightarrow \beta\text{-}Gd_2Ge_3 + L$	1343	Measured	[6]	1343
$\beta\text{-}Gd_2Ge_3 + L \leftrightarrow GdGe_{2.57}$	1163	Measured	[6]	1163
$L \leftrightarrow GdGe_{2.57} + Ge$	1133	Measured	[6]	1133
$GdGe_{2.57} \leftrightarrow \beta\text{-}Gd_2Ge_3 + Ge$	1123	Measured	[6]	1123
$\beta\text{-}Gd_2Ge_3 \leftrightarrow \alpha\text{-}Gd_2Ge_3$	1106	Measured	[6]	1106
$\alpha\text{-}Gd_2Ge_3 + Ge \leftrightarrow \beta\text{-}Gd_5Ge_3$	1081	Measured	[6]	1080
$\beta\text{-}Gd_5Ge_3 \leftrightarrow \alpha\text{-}Gd_5Ge_3$[a]	<873	Evaluated	[5]	872

[a]The allotropic transformation $\beta\text{-}Gd_5Ge_3 \leftrightarrow \alpha\text{-}Gd_5Ge_3$ was evaluated by Gokhale and Abbaschian [5] from the analysis of crystallographic data.

added after optimization in case that the allotropic transformation had been confirmed. Since $\alpha\text{-}Gd_3Ge_5$ is stable at lower temperatures relative to $\beta\text{-}Gd_3Ge_5$, a positive coefficient B is needed for $\alpha\text{-}Gd_3Ge_5$ and it is artificially fixed to be +0.5 due to the lack of experimental data. The coefficient of A for $\alpha\text{-}Gd_3Ge_5$ could be consequently evaluated by calculating the value of $-65{,}305 - 0.5T^*$. T^* is the allotropic transition temperature. It is temporarily assessed to be 872 K in the present work, however, it could be updated according to the future experimental investigations. It is a flexible way to deal with the allotropic transformation.

4.3. Thermodynamic properties

Fig. 3 presents the calculated enthalpies of formation at 298.15 K with the standard element reference state, along with experimental data for comparison. Error bars are plotted according to the reported experimental uncertainties for each piece of experimental data. The present calculation agrees well with the data of Meschel and Kleppa [27], and Usenko et al. [28].

No thermodynamic properties had been reported for liquid. In the present modeling, they were evaluated using the data on liquidus and the equilibria between liquid and solid phases from Eremenko et al. [6]. The presently calculated enthalpies of mixing of liquid were shown in dashed line in Fig. 4. Using the data on liquidus [6], Gokhale and Abbaschian [5] had tentatively assessed the enthalpies of mixing of liquid. A maximum of 110.45 kJ/mol at x(Ge) was estimated [5], which is nearly twice as large as that evaluated in this modeling. The present calculation was expected to be a much more reasonable prediction, since it was evaluated based on a thorough thermodynamic modeling, taking into account all thermodynamic properties and phase equilibria data, including the data on liquidus [6].

As shown in solid line in Fig. 4, the calculated excess entropy of mixing of liquid is quite reasonable. It shows negative values in the entire composition range and exhibits a minimum at x(Ge) = 0.5, which is no less than −2.2 J/mol.

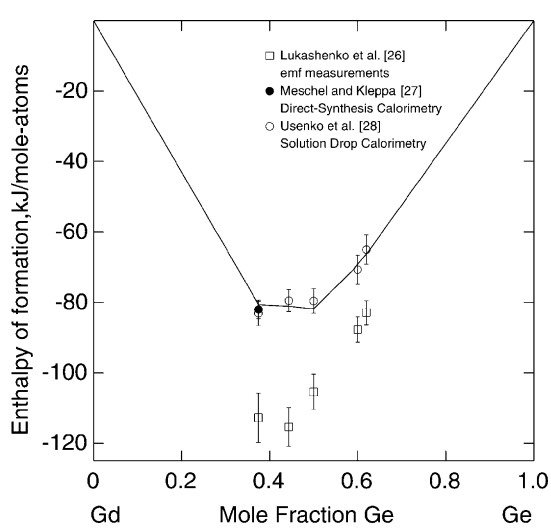

Fig. 3. Calculated standard enthalpies of formation of solid phases at 298.15 K, compared with experimental data [26–28]. The data in Ref. [26] was not used in the optimization.

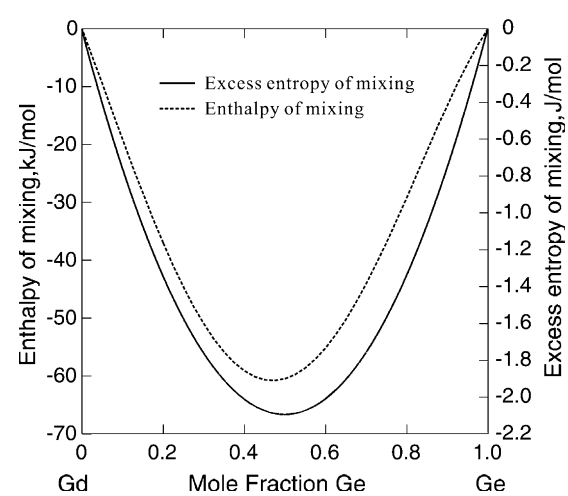

Fig. 4. Calculated enthalpies of mixing and excess entropies of mixing of liquid, which are independent of temperature.

5. Conclusions

A thermodynamic modeling of the Gd–Ge system is carried out. The comprehensive comparisons show that the calculated phase diagram and thermodynamic properties agree well with the accepted experimental data. The thermodynamic description of pure Gd from Shen et al. [1] is modified to well describe the allotropic transition α-Gd ↔ β-Gd.

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Appendix A

The thermodynamic description of Gd from Shen et al. [1]

β-Gd (bcc-A2)

$$
{}^{0}G_{\mathrm{Gd}}^{\mathrm{bcc\text{-}A2}} - H^{\mathrm{SER}} = \begin{cases} -3600.77684 + 95.01916415T - 24.7214131T \ln T - .00285240521T^2 - 3.14674076\mathrm{E}-07T^3 - 8665.73348T^{-1} & 200.00 < T < 1000.00 \\ \underline{+152792.743} - 1349.58873T + 180.097094T \ln T - .119550229T^2 + 1.17915728\mathrm{E}-05T^3 - 22038836.7T^{-1} & 1000.00 < T < 1508.15 \\ \underline{-15783.7618} + 202.222057T - 38.960425T \ln T & 1508.15 < T < 1586.15 \\ -19850.5562 + 224.817909T - 41.904333T \ln T + 8.58222759\mathrm{E}-04T^2 - 3.77570269\mathrm{E}-08T^3 + 995428.573T^{-1} & 1586.15 < T < 3600.00 \end{cases}
$$

α-Gd (hcp-A3) ($T_c = 293.4$, $B_o = 3.0$)

$$
{}^{0}G_{\mathrm{Gd}}^{\mathrm{hcp-A3}} - H^{\mathrm{SER}} = \begin{cases} -6834.5855 + 97.13101T - 24.7214131T \ln T - .00285240521T^2 - 3.14674076\mathrm{E}-07T^3 - 8665.73348T^{-1} & 200.00 < T < 1000.00 \\ -6483.25362 + 95.6919924T - 24.6598297T \ln T - .00185225011T^2 - 6.61211607\mathrm{E}-07T^3 & 1000.00 < T < 1508.15 \\ -123124.992 + 699.125537T - 101.800197T \ln T + .0150644246T^2 - 6.39165948\mathrm{E}-07T^3 + 29356890.3T^{-1} & 1508.15 < T < 3600.00 \end{cases}
$$

Liquid

$$
{}^{0}G_{\mathrm{Gd}}^{\mathrm{liq}} - H^{\mathrm{SER}} = \begin{cases} +6225.4407 + 88.80921026T - 24.7214131T \ln T - .00285240521T^2 - 3.14674076\mathrm{E}-07T^3 - 8665.73348T^{-1} & 200.00 < T < 1000.00 \\ +146262.037 - 1208.70685T + 159.352082T \ln T - 0.108247135T^2 + 1.06945505\mathrm{E}-05T^3 - 19678357T^{-1} & 1000.00 < T < 1508.15 \\ -5397.314 + 192.336215T - 38.5075T \ln T & 1508.15 < T < 3600.00 \end{cases}
$$

Note: Gibbs energy in J/mol and T in Kelvin. In order to well describe the allotropic transition α-Gd ↔ β-Gd, the thermodynamic description of β-Gd is presently modified by either substituting −15783.5818 for the underlined term of −15783.7618 (Modification A) or substituting +152792.733 for the underlined term of +152792.743 (Modification B).

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